BORGWARNER

Cascadia Motion Calibration Manual

Document: 0A-0166-01

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1 Safety First

	When you see this symbol, PAY ATTENTION. This indicates that something important is about to be said, that concerns your safety and the proper operation of the equipment.
	When you see this symbol, you are being alerted to an IMMEDIATE DANGER that could cause severe injury or even death. You MUST review these sections carefully and do everything possible to comply with installation and operation requirements, or you risk injury or death to yourself or anyone else who uses the equipment or the vehicle. Failure to comply with safety requirements will void all warranties and could expose you as the installer to liability in the event of an injury. Use the equipment in the manner in which it was intended.
	When you see this symbol, you are being advised that the issue under discussion has a serious safety or equipment reliability implication. Use caution and be conservative. Use equipment in the manner described in this Hardware User's Manual.

Safety is entirely the responsibility of the installer of this equipment. BorgWarner Portland (formerly Cascadia Motion) has done everything it can to ensure that the traction inverter itself conforms to international standards for safety. This does NOT mean that your installation will be safe, or that it will not interfere with other systems on board your vehicle. It is your responsibility as the installer to review this entire Hardware User's Manual, to understand the implications of each and every section, and to know what might be unique about your system application that then presents a unique hazard or potential safety issue – and to solve it.

2 Overview

Before the inverter can be used successfully, it is very important to make sure that it is calibrated properly. There are several calibrations that are performed before each unit is shipped to the customer. However, some of these calibrations depend on the specific environment in which the unit is used.

User Manuals for following calibration processes are provided to customers. The calibrations can be performed as many times as needed.

Table 1: Calibration Process

Calibration Process	**User Manual (PDF format)**	**Factory Calibrated?**
Current Offset	Section 5, 'Current Offset Calibration	No, not necessary
DC Voltage	Section 6, 'DC Voltage Calibration (factory calibrated thus not normally needed)	Yes
Resolver	See 'Cascadia Motion SW User Manual' (this process is necessary for all motors that use a resolver or SIN/COS encoder)	No
SIN/COS Encoder	Section 8, 'Encoder Calibration' (only necessary with certain motors that have a sin/cos encoder)	No
Hall Sensor Encoder	Section 9, 'Hall Sensor Calibration (not normally needed)	No
RTD	Section 10, 'RTD Calibration' (factory calibrated thus not normally needed)	Yes

3 Factory Calibration Restoration

This section describes how a CM series inverter's calibration data can be stored and restored into the inverter's non-volatile memory. The inverter must be running software version 652C or newer. If the inverter left the factory with 652C or newer, then the inverter's calibration data has already been stored and can either be updated or retrieved. This document outlines how to check if a calibration data backup is already present via a calibration restoration defsyms file.

If the inverter's hardware type is reported as 37750, please contact Cascadia Motion support.

3.1 Parameters Protected

Most of the EEPROM parameters protected by the calibration restoration feature are not typically made available to the user to change. Below is an exhaustive list of the EEPROM parameters protected by the calibration restoration feature.

- `HW_Version_EEPROM`
- `Serial_Number_EEPROM`
- `VDC_ADC_MAX_EEPROM_(V)_x_10`
- `VDC_OFFSET_EEPROM_(V)_x_10`
- `VAB_ADC_MAX_EEPROM_(V)_x_10`
- `VAB_OFFSET_EEPROM_(V)_x_10`
- `IA_ADC_MAX_EEPROM_(A)_x_10`
- `IB_ADC_MAX_EEPROM_(A)_x_10`
- `IC_ADC_MAX_EEPROM_(A)_x_10`
- `VBC_ADC_MAX_EEPROM_(V)_x_10`
- `VBC_OFFSET_EEPROM_(V)_x_10`
- `VDC_SDFM_OFFSET_EEPROM_(V)_x_10`
- `VA_SDFM_OFFSET_EEPROM_(V)_x_10`
- `VB_SDFM_OFFSET_EEPROM_(V)_x_10`
- `VC_SDFM_OFFSET_EEPROM_(V)_x_10`
- `VDC_SDFM_MAX_EEPROM_(V)_x_10`
- `VA_SDFM_MAX_EEPROM_(V)_x_10`
- `VB_SDFM_MAX_EEPROM_(V)_x_10`
- `VC_SDFM_MAX_EEPROM_(V)_x_10`
- `IDC_ADC_MAX_EEPROM_(A)_x_10`
- `IDC_OFFSET_EEPROM_(A)_x_10`
- `L100_RTD1_100_OHM_SQR_GAIN_EEPROM`
- `L100_RTD1_100_OHM_GAIN_EEPROM`
- `L100_RTD1_100_OHM_OFFSET_EEPROM`
- `L100_RTD2_100_OHM_SQR_GAIN_EEPROM`
- `L100_RTD2_100_OHM_GAIN_EEPROM`
- `L100_RTD2_100_OHM_OFFSET_EEPROM`
- `L100_RTD1_1K_OHM_SQR_GAIN_EEPROM`
- `L100_RTD1_1K_OHM_GAIN_EEPROM`
- `L100_RTD1_1K_OHM_OFFSET_EEPROM`
- `L100_RTD2_1K_OHM_SQR_GAIN_EEPROM`
- `L100_RTD2_1K_OHM_GAIN_EEPROM`
- `L100_RTD2_1K_OHM_OFFSET_EEPROM`

3.2 Procedure: Calibration Restoration

In the case where a user has upgraded their firmware from a software version before 652C, to 652C or newer, they can manually backup their inverter's calibration data if desired. This section serves as a walkthrough on how to do so.

1. Launch the RMS GUI and load the 'defsyms_eeprom_cal_restore.txt' file. See figure 1 for details.
2. Add the `CAL_Backup_Status` to the Watch List. Verify it reads 1, which corresponds to calibration present.
3. Add the `EEprom_Load_Cal_Command` to the Command List.
4. Change the value from 0 to 1. The value should change back to 0 if the command is sent correctly.
5. Verify the `CAL_Backup_Status` value has been updated to 2, indicating calibration loaded.

4 HW Version Update Process

HW Version (`HW_Version_EEPROM`) is an EEPROM parameter that tells the controller what type of hardware is actually in place. This allows the controller to distinguish between a CM200DX and CM200DZ, for example. It also can be used to modify certain operating limits.

This parameter is factory preset and normally should not require change.

4.1 Procedure

Here are the steps you need to follow to change the HW version:

1. The file 'defsyms_HW_version_mod.txt' should have been provided with this document. Save it in the same folder where the firmware (same firmware that is using on the inverter to be modified) and the existing GUI defsyms file is located.
2. Make sure that computer is connected to the inverter and 12V power is applied.
3. Start the application RMS GUI.
4. From the 'File' menu, select 'Load Symbols File' and select 'defsyms_HW_version_mod.txt'.
5. Click on the tab, EEPROM View.
6. Click on 'Save' button to save the values relevant to HW Version.
7. Click on the value of the parameter, `HW_Version_EEPROM` and change it to the desired new value. Important Note: This is an important parameter that must not be changed without careful consideration and consultation with Cascadia Motion.
8. Click on 'Program EEPROM' button. You should see EEPROM Programming Successful message. Cycle power to the inverter. Close the RMS GUI application.
9. Restart the RMS GUI. The hardware version has been changed. The title bar of the RMS GUI application should reflect the new hardware version.
10. Click on 'Save' button to save the complete set of current values in the defsyms file.

5 Current Offset Calibration

Contact Cascadia Motion support for additional information.

6 DC Voltage Calibration

6.1 Procedure

1. Apply 12-V power to the inverter.
2. Launch the RMS GUI and load the 'defsyms_vdc_cal.txt' file. See figure 1 for details.
3. Apply 400V to the inverter's HVDC bus if it's a DX, 600V if it's a DZ.
4. Measure the actual voltage using a volt-meter and enter it in the Excel spreadsheet *VDC Calibration.xlsx*
5. Look up the voltage shown in 'Memory View' of the GUI and enter it in the second line of the spreadsheet. This completes data for the first set point.
6. Now, change the voltage to 250V for DX inverters, 400V for DZ.
7. Repeat steps 4 and 5 to collect data for set point 2 and enter it in the Excel spreadsheet lines 3 and 4.
8. Enter the values for `VDC_ADC_MAX_EEPROM_(V)_x_10` and `VDC_OFFSET_EEPROM_(V)_x_10` in the spreadsheet, respectively.
9. New values for `VDC_ADC_MAX_EEPROM_(V)_x_10` and `VDC_OFFSET_EEPROM_(V)_x_10` should be automatically calculated.
10. Enter these new values in the GUI and press the 'Program EEPROM' button.

7 Resolver Calibration

See 'Cascadia Motion SW User Manual' for details.

8 Encoder Calibration

This section describes the process to calibrate a SIN/COS encoder to work with Cascadia Motion inverters. The procedure finds the offset voltage of the encoder.

This process applies to CMxxx inverters that have the option -SP (e.g. CM200DZ-SP).

This process is most often used with Yasa motors.

The SIN/COS encoder inputs are single ended analog signals that provide a signal that varies with the motor position. The signals are similar to what a resolver provides but don't have the excitation carrier waveform.

The SIN/COS signal should meet the following requirements:

Offset Voltage	2.5V ± 0.2V
Signal Voltage	3.8V pk-pk maximum

For CMxxx inverters the SIN/COS input impedance is 10K ohm to 1.9V.

For the CMxxx inverters (48 pin Molex CMC I/O connector) the SIN/COS connections for the inverter are:

Encoder Signal	Pin #
SIN (normally ENC_A)	H1
COS (normally ENC_B)	J1
Encoder Power (5V)	J2
Encoder GND	K2
Encoder Shield	G1

For CMxxx inverters the inverter must be an -SP configuration. At a future data CMxxx inverters (starting with the CM350SiC) will be able to automatically reconfigure themselves to be a -SP option.

8.1 Procedure

The purpose of this calibration process is to increase the accuracy of sin and cosine inputs to the encoder. There may be an offset associated with each input based on minimum and maximum values.

Important Note: It is not necessary for the high-voltage DC to be connected to the inverter. However, spinning the motor will generate a DC voltage in the inverter, even if the inverter is off. This voltage is dangerous, take proper precautions. If a high voltage DC is connected to the inverter make sure that it is high enough in value that the voltage generated by the motor will be less than this battery voltage.

The voltage that the inverter is seeing at the SIN/COS inputs can be viewed with the RMS GUI using the following two Memory Window parameters:

Parameter	Description
`Sin_corr_(V)(x100)`	Value of the SIN input in volts.
`Cos_corr_(V)(x100)`	Value of the COS input in volts.

For some versions of code the above two parameters may not show in the Memory window. If they do not show then open the particular defsyms file that the GUI is using in a text editor. Look for these two lines to see if they are proceeded by a "#". If they are then remove the "#" and then save and close the file. Restart the RMS GUI and the two lines should appear.

Monitoring of these two signals can help make sure that as the motor rotates the two signals are varying in voltage. They should be 90 degrees offset from each other.

In order to perform the calibration process, follow these steps:

1. Start the RMS GUI application.
2. On the EEPROM page of the GUI, make sure that the correct motor type is selected. If not, then select the correct motor type and program EEPROM parameters. The inverter power needs to be recycled after EEPROM parameter programming.
3. Add the following parameters to the Watch Window of the GUI (they are located at the bottom of the list of parameters):

GUI Parameter	Default Value
`Encoder_Calibration_Command`	0 (Calibration not in progress)
`Sin_Minimum_Value_(V)(x100)`	0 (0.00 V)
`Sin_Maximum_Value_(V)(x100)`	500 (5.00 V)
`Sin_Offset_(V)(x100)`	250 (2.50 V)
`Cos_Minimum_Value_(V)(x100)`	0 (0.00 V)
`Cos_Maximum_Value_(V)(x100)`	500 (5.00 V)
`Cos_Offset_(V)(x100)`	250 (2.50 V)

4. The calibration process can be performed only if all of the following conditions are true:
   (a) Correct Motor Type is selected
   (b) Inverter is disabled
   (c) VSM State is VSM_WAIT (4)
   (d) There are no POST or RUN faults present
5. Enter a '1' (Perform Calibration) command for `Encoder_Calibration_Command` in the GUI. Make sure the Auto Continuous Refresh is off on the GUI when changing the command. `Encoder_Calibration_Command` can only have the following values:
   (a) 0: Calibration not in progress
   (b) 1: Perform Calibration

(c) 2: Finish Calibration

6. Turn the Auto Continuous Refresh back on to monitor the values. Rotate the motor slowly by hand. Observe `Sin_Minimum_Value_(V)(x100)` and `Sin_Maximum_Value_(V)(x100)` changing. Eventually, `Sin_Minimum_Value_(V)(x100)` and `Sin_Maximum_Value_(V)(x100)` will show the minimum and maximum values, respectively. Same is true for `Cos_Minimum_Value_(V)(x100)` and `Cos_Minimum_Value_(V)(x100)`. Rotate through several revolutions.

7. Turn off the Auto Continuous Refresh. Enter a '2' (Finish Calibration) command to complete the process. `Encoder_Calibration_Command` will change back to '0' to show that the unit is not in calibration process any more. Also, `Sin_Offset_(V)(x100)` and `Cos_Offset_(V)(x100)` will be calculated and saved in EEPROM for future use. Turn the 12V power off and on. Verify that the EEPROM values have been updated. The EEPROM values can be viewed with following parameters:

RMS GUI Parameter	Default Value
`Sin_Offset_EEPROM_(V)_x_100`	250
`Cos_Offset_EEPROM_(V)_x_100`	250

Typically these two parameters should not vary substantially from the default values.

8.2 Gamma Adjust

Finally, it is necessary to perform the gamma adjust on the Sin/Cos encoder. Please see section Resolver Calibration. This section outlines determining the amount of gamma adjust needed for the motor and inverter pair. Gamma Adjust corrects for the angle offset between the mechanical position of the sensor and the magnet position.

NOTE: The gamma adjust is a required step and cannot be bypassed.

9 Hall Sensor Calibration

Contact Cascadia Motion support for additional information.

10 RTD Calibration

Contact Cascadia Motion support for additional information.

A Appendix: EEPROM Configuration Parameters

EEPROM Parameter	GUI Addr.	CAN Addr.	Value Range	Description
Serial_Number_EEPROM	TBD	TBD	Internal	The inverter's serial number.
Manufacturing_ID_1_EEPROM	0x0176	0x0111	Internal	Manufacturing ID bytes 0,1.
Manufacturing_ID_2_EEPROM	0x017A	0x0112	Internal	Manufacturing ID bytes 2,3.
Manufacturing_ID_3_EEPROM	0x01C2	0x0113	Internal	Manufacturing ID bytes 4,5.

Table 2: Serial Number EEPROMs

EEPROM Parameter	GUI Addr.	CAN Addr.	Value Range	Description
Gamma_Adjust_EEPROM_(Deg)_x_10	TBD	TBD	0 – 65535	Parameter used for encoder calibration.
Sin_Offset_EEPROM_(V)_x_100	TBD	TBD	0 – 5000	Parameter used for encoder calibration.
Cos_Offset_EEPROM_(V)_x_100	TBD	TBD	0 – 5000	Parameter used for encoder calibration.
HW_Version_EEPROM	TBD	TBD	Internal	The inverter's hardware version.
VDC_ADC_MAX_EEPROM_(V)_x_10	0x0141	0x00C2	0 – 65535	The inverter's voltage calibration maximum.
VDC_OFFSET_EEPROM_(V)_x_10	0x0142	0x00C3	-32768 – 32767	The inverter's voltage calibration offset.
VAB_ADC_MAX_EEPROM_(V)_x_10	0x0143	0x00C4	0 – 65535	The inverter's phase voltage measurement max. Not available on all inverters.
VAB_OFFSET_EEPROM_(V)_x_10	0x0144	0x00C5	-32768 – 32767	The inverter's phase voltage measurement max. Not available on all inverters.
VBC_ADC_MAX_EEPROM_(V)_x_10	0x0153	TBD	0 – 65535	The inverter's phase voltage measurement max. Not available on all inverters.
VBC_OFFSET_EEPROM_(V)_x_10	0x0154	TBD	-32768 – 32767	The inverter's phase voltage measurement offset. Not available on all inverters.
VDC_SDFM_MAX_EEPROM_(V)_x_10	0x0165	TBD	0 – 65535	The inverter's DC bus voltage measurement maximum. Not available on all inverters.
VDC_SDFM_OFFSET_EEPROM_(V)_x_10	0x0166	TBD	-32768 – 32767	The inverter's DC bus voltage measurement maximum. Not available on all inverters.

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EEPROM Parameter	GUI Addr.	CAN Addr.	Value Range	Description
VA_SDFM_MAX_EEPROM_(V)_x_10	0x0167	TBD	0 – 65535	The inverter's phase voltage measurement max. Not available on all inverters.
VA_SDFM_OFFSET_EEPROM_(V)_x_10	0x0168	TBD	-32768 – 32767	The inverter's phase voltage measurement offset. Not available on all inverters.
VB_SDFM_MAX_EEPROM_(V)_x_10	0x0169	TBD	0 – 65535	The inverter's phase voltage measurement max. Not available on all inverters.
VB_SDFM_OFFSET_EEPROM_(V)_x_10	0x016A	TBD	-32768 – 32767	The inverter's phase voltage measurement offset. Not available on all inverters.
VC_SDFM_MAX_EEPROM_(V)_x_10	0x016B	TBD	0 – 65535	The inverter's phase voltage measurement max. Not available on all inverters.
VC_SDFM_OFFSET_EEPROM_(V)_x_10	0x016C	TBD	-32768 – 32767	The inverter's phase voltage measurement offset. Not available on all inverters.

Table 3: Voltage Sensor Calibration EEPROMs

EEPROM Parameter	GUI Addr.	CAN Addr.	Value Range	Description
IDC_ADC_MAX_EEPROM_(A)_x_10	0x0146	TBD	0 – 65535	The inverter's DC current measurement max. Not available on all inverters.
IDC_OFFSET_EEPROM_(A)_x_10	0x0147	TBD	-32768 – 32767	The inverter's DC current measurement offset. Not available on all inverters.
IA_ADC_MAX_EEPROM_(A)_x_10	0x01B1	TBD	0 – 65535	The inverter's phase current measurement max.
IB_ADC_MAX_EEPROM_(A)_x_10	0x01B2	TBD	0 – 65535	The inverter's phase current measurement max.
IC_ADC_MAX_EEPROM_(A)_x_10	0x01B3	TBD	0 – 65535	The inverter's phase current measurement max.

Table 4: Current Sensor Calibration EEPROMs

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EEPROM Parameter	GUI Addr.	CAN Addr.	Value Range	Description
RTD_Selection_EEPROM_(BITS_1_0)	0x014A	TBD	0 – 3	0: RTD1 = PT1000 RTD2 = PT1000 1: RTD1 = PT100 RTD2 = PT1000 2: RTD1 = PT1000 RTD2 = PT100 3: RTD1 = PT100 RTD2 = PT100
L100_RTD1_100_OHM_SQR_GAIN_EEPROM	0x01B4	NA	TBD	TBD
L100_RTD1_100_OHM_GAIN_EEPROM	0x0155	NA	TBD	TBD
L100_RTD1_100_OHM_OFFSET_EEPROM	0x0156	NA	TBD	TBD
L100_RTD2_100_OHM_SQR_GAIN_EEPROM	0x01B6	NA	TBD	TBD
L100_RTD2_100_OHM_GAIN_EEPROM	0x0159	NA	TBD	TBD
L100_RTD2_100_OHM_OFFSET_EEPROM	0x015A	NA	TBD	TBD
L100_RTD1_1K_OHM_SQR_GAIN_EEPROM	0x01B5	NA	TBD	TBD
L100_RTD1_1K_OHM_GAIN_EEPROM	0x0157	NA	TBD	TBD
L100_RTD1_1K_OHM_OFFSET_EEPROM	0x015B	NA	TBD	TBD
L100_RTD2_1K_OHM_SQR_GAIN_EEPROM	0x01B7	NA	TBD	TBD
L100_RTD2_1K_OHM_GAIN_EEPROM	0x015B	NA	TBD	TBD
L100_RTD2_1K_OHM_OFFSET_EEPROM	0x015C	NA	TBD	TBD

Table 5: RTD Calibration EEPROMs

B Appendix: GUI Parameters

B.1 GUI Watch Parameters

EEPROM Parameter	GUI Addr.	CAN Addr.	Description
`Gamma_Adjust_(Deg)_x_10`	TBD	TBD	Parameter used for encoder calibration.
`Delta_Resolver_In_Fil_(Deg)_x_10`	TBD	TBD	
`Sin_corr_(V)(x100)`	TBD	TBD	
`Cos_corr_(V)(x100)`	TBD	TBD	
`Cos_Offset_(V)(x100)`	TBD	TBD	
`Sin_Offset_(V)(x100)`	TBD	TBD	
`Sin_Minimum_Value_(V)(x100)`	TBD	TBD	
`Sin_Maximum_Value_(V)(x100)`	TBD	TBD	
`Cos_Minimum_Value_(V)(x100)`	TBD	TBD	
`Cos_Maximum_Value_(V)(x100)`	TBD	TBD	
`CAL_Backup_Status`	TBD	TBD	Calibration backup status.

B.2 GUI Command Parameters

EEPROM Parameter	GUI Addr.	CAN Addr.	Description
`EEprom_Load_Cal_Command`	0x0005	0x0047	Non-zero values written to this parameter will trigger an EEPROM calibration backup load.
`EEprom_Backup_Cal_Command`	0x0025	0x0046	Non-zero values written to this parameter will trigger an EEPROM calibration backup store. Note that this will overwrite factory settings. See section 3, 'Factory Calibration Restoration'.
`RTD_Select_Command_(BITS_1_0)`	0x0014	NA	RTD Selection command bits for usage in RTD calibration.
`Encoder_Calibration_Command`	TBD	TBD	TBD

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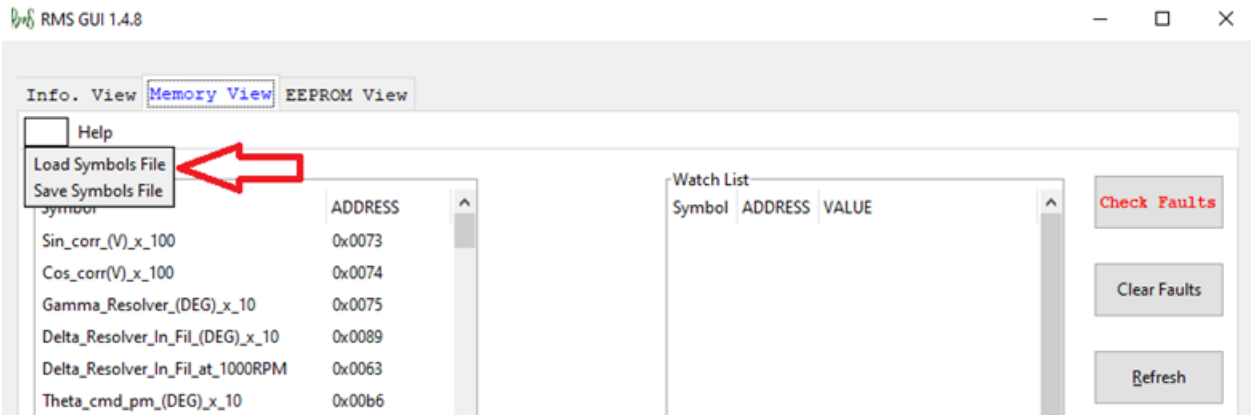

Figure 1: Defsyms file loading

C Appendix: Loading a Defsyms File

C.1 Procedure

1. Apply 12-V power to the inverter.
2. Start the RMS GUI application from the firmware folder.
3. Click the 'File...' menu and select 'Load Symbols File'.

D Revision History

Version	Description	Updated By	Date
0A-0166-01	Initial Content, ECO 1142	Noah Erickson	04/22/2025